

Refresh Opportunity Scoring using Machine Learning

FlyRank ML Internship Capstone

Lane: Refresh / Content Opportunity Scoring

Author: Aditya S Ram

Organization: FlyRank AI Internship

Abstract

This project develops a machine learning framework to identify web pages that are likely to require content refresh. Historical search performance data from the FlyRank internship warehouse is transformed into page-level features including impressions, clicks, ranking position, CTR, and growth metrics. Multiple machine learning algorithms are trained and compared to estimate refresh likelihood. Feature importance analysis identifies the strongest indicators of declining search performance. The resulting ranked recommendations can help prioritize SEO content updates.

Introduction

Large websites often contain thousands of pages, making manual identification of declining content impractical.

The objective of this project is to develop an automated Refresh Opportunity Scoring system that identifies pages experiencing declining search performance.

The generated rankings support data-driven content optimization decisions.

Dataset

Dataset Source

FlyRank ML Internship Warehouse

Warehouse Size

Approximately 79 million daily records

Filtered Dataset

111,033 content pages

Features

Impressions

Clicks

Average Position

CTR

Growth Metrics

Position Change

Methodology

Workflow

1. Connect to FlyRank Warehouse
2. Aggregate page-level metrics
3. Engineer new features
4. Train Machine Learning models
5. Compare model performance
6. Rank refresh opportunities
7. Export recommendations

Results

Model	Accuracy
Logistic Regression	99.996%
Decision Tree	100.0%
Random Forest	100.0%
Gradient Boosting	100.0%

The Random Forest model achieved the strongest performance among the evaluated models.

Feature importance analysis showed that impression growth, click growth, and previous impressions were the most influential predictors for identifying refresh opportunities.

Discussion

The machine learning models successfully identified pages showing patterns associated with declining search performance.

Among all evaluated models, the Random Forest classifier achieved the strongest performance while providing interpretable feature importance rankings.

The analysis indicates that impression trends, click behavior, and ranking position are the strongest indicators of pages requiring review.

These findings demonstrate that multiple search signals can be combined to prioritize content refresh decisions more effectively than manual inspection.

Limitations

The current implementation defines the target label using historical performance changes.

As a result, some input features are closely related to the target, which may inflate evaluation metrics.

Future work should define prediction targets using future performance windows and perform time-aware validation to reduce leakage.

The model should be interpreted as a decision-support system rather than proof that refreshing a page will improve search rankings.

Recommendations

The generated ranking can be used by SEO editors to prioritize content updates.

Recommended actions include:

- Refresh outdated information.
- Improve titles and metadata.
- Expand thin content.
- Improve internal linking.
- Monitor page performance after publishing updates.

Higher-ranked pages should be reviewed first because they show stronger indicators of declining search performance.

Conclusion

This project demonstrates an end-to-end machine learning workflow for identifying refresh opportunities within large-scale search performance datasets.

The developed ranking model provides a scalable approach for prioritizing SEO content maintenance.

Future work will include leakage-free target construction, additional ranking signals, and time-aware evaluation to improve model robustness.

Acknowledgments

Built on the FlyRank ML Internship dataset.

Data Source: <https://flyrank.ai>